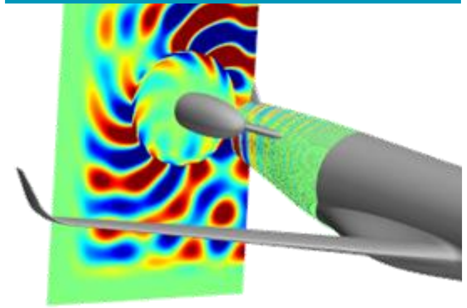
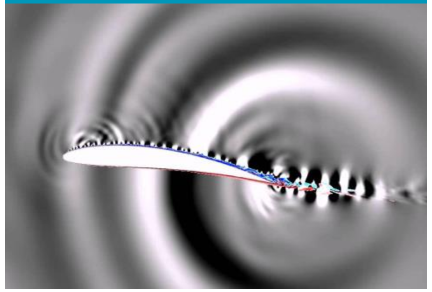
Actran for Aircraft Exterior Noise and Interior Noise

Applications in Aircraft

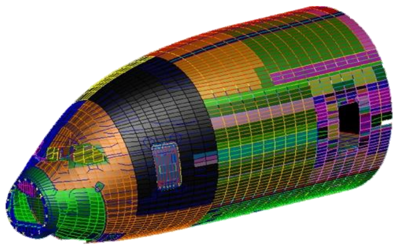
Exterior noise: Propeller



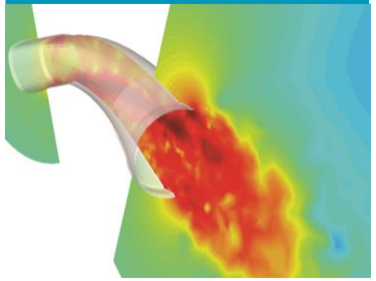
Exterior Airframe Noise (Landing gears, wings…)



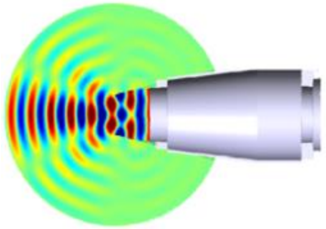
Interior Noise: Transmission through Fuselage



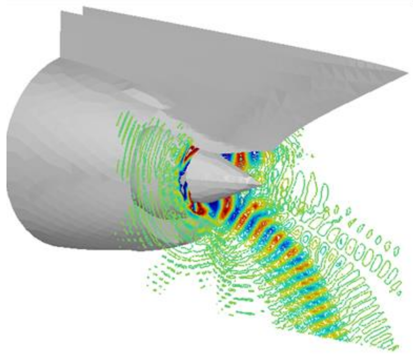
Interior noise: Air Conditioning System



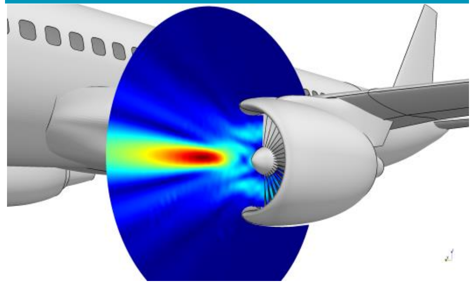
Exterior noise: Auxiliary Power Unit (APU)



Exterior noise: Engine Nacelle Exhaust



Exterior noise: Engine Nacelle Intake



Summary

1. Exterior noise
    - Acoustic propagation for Air Intake and By-pass
    - Acoustic propagation for Exhaust and CROR Engines
    - Ramp noise
    - Aeroacoustic simulations

2. Interior noise
    - Turbulent boundary layer on fuselage and windshield
    - Engines (intake, exhaust, CROR engines)
    - Air conditioning

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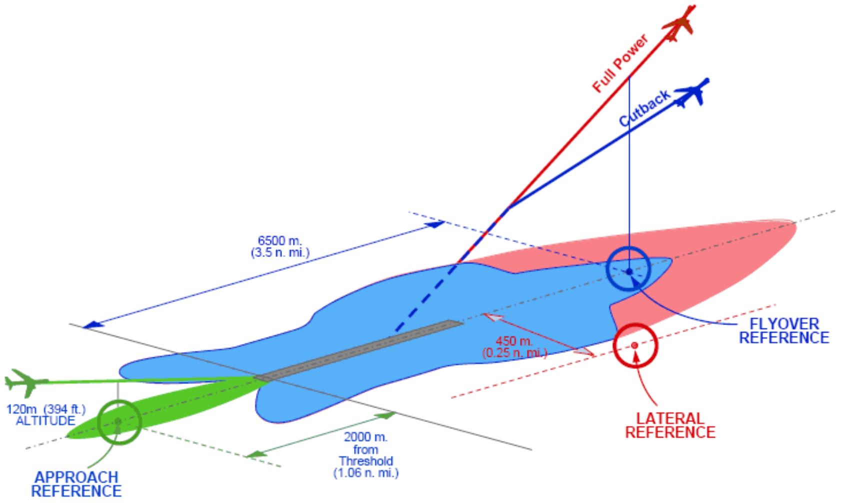
Aircraft Noise Certification

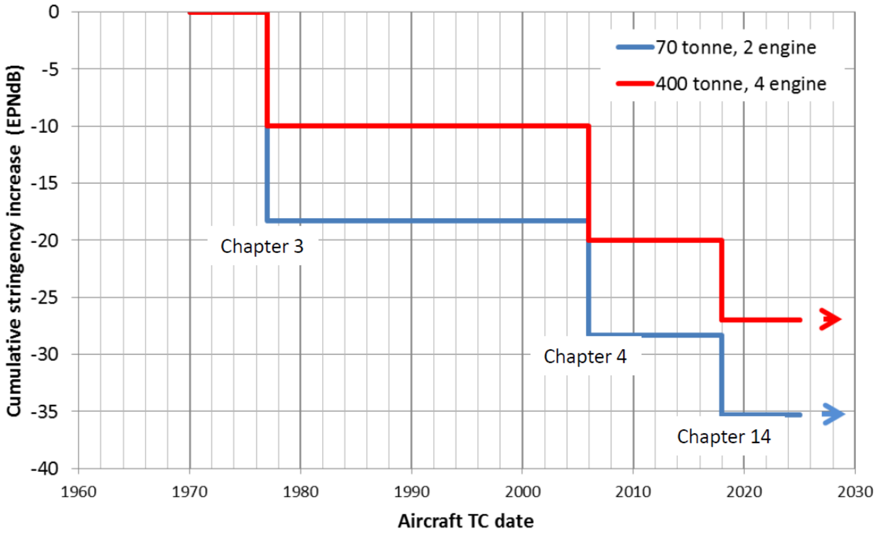
Aircraft Noise Requirements

- International Regulations (FAR Part 36) and local airport requirements (e.g. London City Airport):
  - Prescribe three certification conditions: Approach, Flyover, Sideline.

- Regulations are becoming more and more restrictive
  - An aircraft shall be certified well below the limit otherwise it could become obsolete over the years



Aircraft Certification Conditions



Evolution of Aircraft Noise Limits

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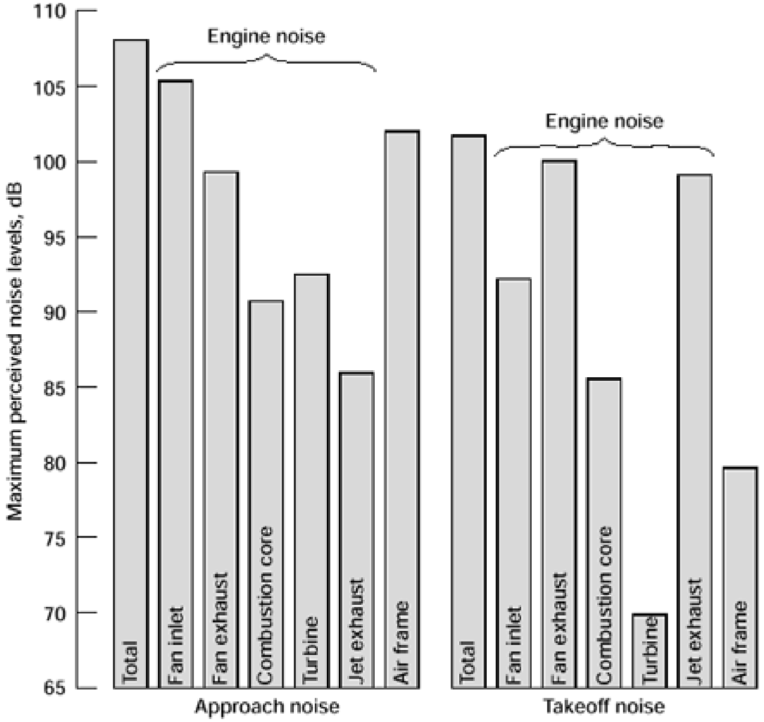
Aircraft Exterior Noise

- In all certification conditions, engine noise is the most relevant source



Source: adg.stanford.edu

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Turbo Engine Noise and Regulations

- Turbo engine: Major source of noise for the resident during take-off and landing
  - Economical issue (night flying restriction, noise tax,…)
  - Health issue (stress level, heart disease,…) - Harvard School of Public Health, 2013
  - Political issue (landing and take-off route)

- Maximum level of noise ruled by national legislation

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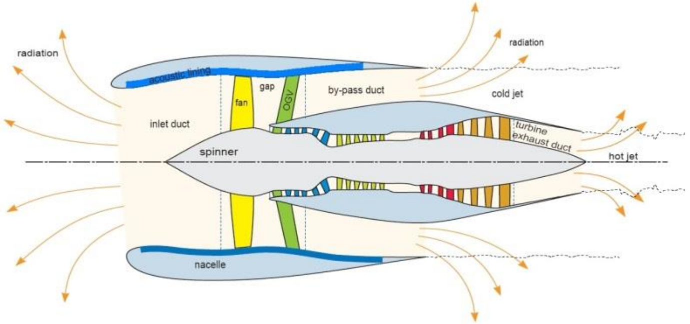
Air Intake and By-pass duct design

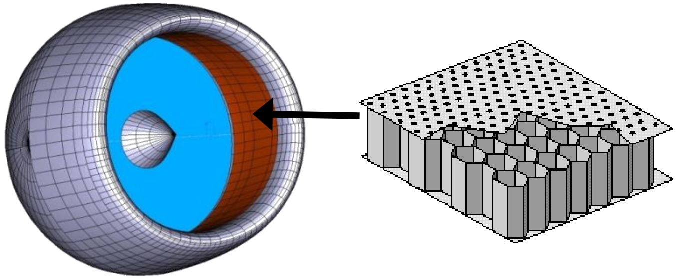
Air Intake and By-pass duct

- Noise due to:
  - The fan and rotor-stator interactions (mainly)
- How to reduce noise:
  - Use acoustic treatments (liners)
- With Actran you can:
  - Model easily noise sources thanks to duct modes
  - Perform convected acoustic propagation
  - Import mean flows from many CFD formats
  - Model acoustic liners with admittance condition or with cavities and perforated plates
  - Evaluate IL and TL, the reflected power, the dissipated power and the radiated power
  - Use acoustic non-reflecting boundary conditions in an homogeneous convected media: infinite elements and (A)PML or with free duct modes

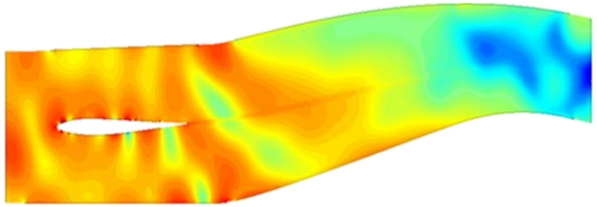
Turbo fan engine:
inlet by-pass and exhaust



Acoustic treatment modeling



Pressure field in a by-pass duct



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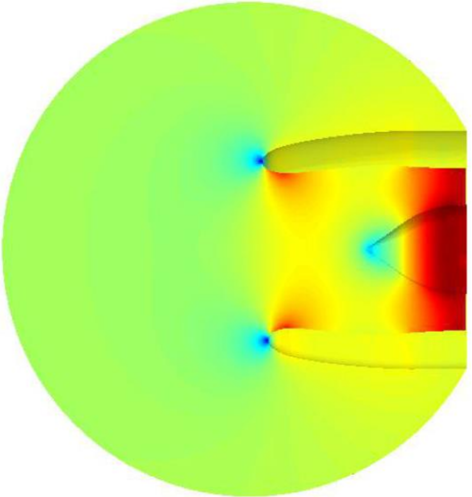
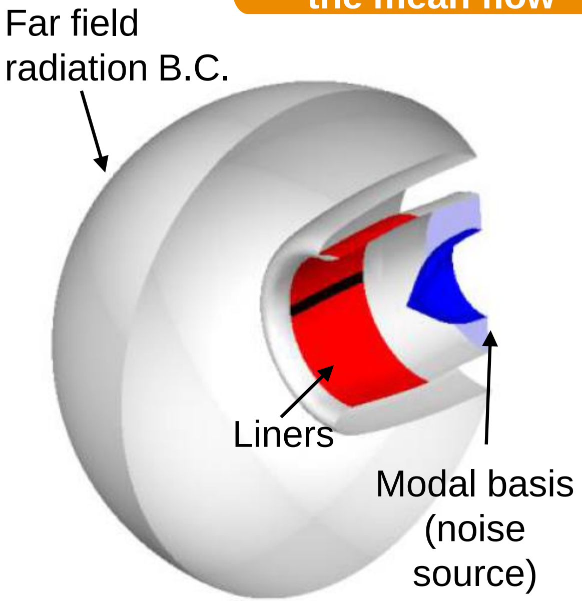
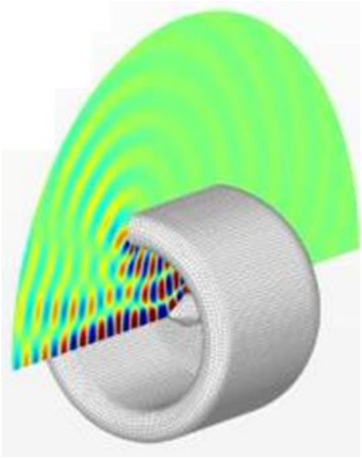
Process for Air Intake Acoustic Radiation

Acoustic model

- **Acoustic Liners**
- **Modal basis**
- **Interpolation of the mean flow**







Mean Flow

- **Actran compressible flow solver or imported from an external code**

Acoustic Results

- **SPL in Near and far field in presence of mean flow**
- **Radiated and absorbed power**

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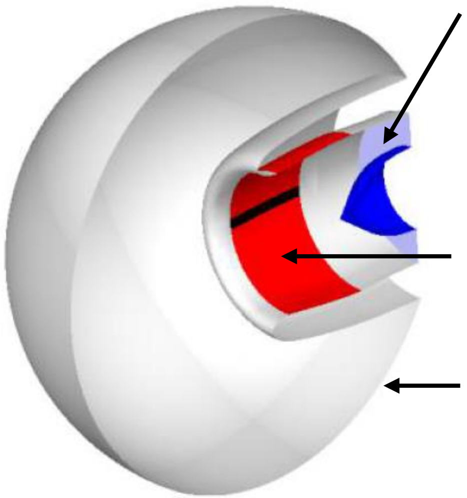
Typical By-pass and Air Intake Models

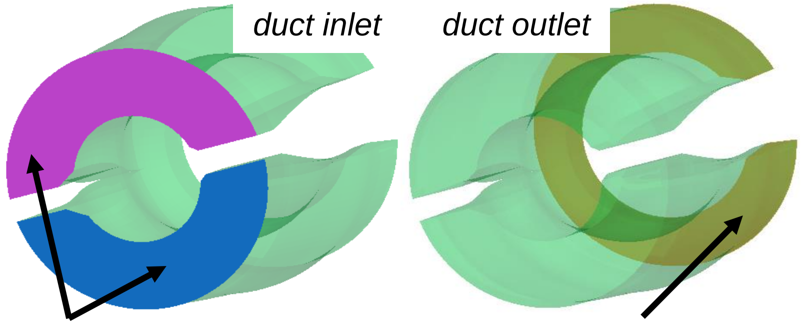
Key ingredients

- Acoustic treatments are modeled with surface admittance:
  - Inside the air intake
  - Inside the by-pass duct
- Acoustic indicators calculated:
  - Air intake: Sound pressure level in near and far field
  - By-pass duct: Transmission Loss

- Modal bases are used to model:
  - The noise source
  - The Non Reflecting Boundary Condition (NRBC) in ducts which model a semi-infinite duct
- Analytical duct modes are used for:
  - Annular or circular duct
- Numerical duct modes are use for:
  - Arbitrary duct



Typical Air Intake Model



Typical By-Pass Duct Model

AIRBUS

Silent but powerful aicraft engine

"Airbus has confirmed the accuracy of Actran predictions by comparing them with engine static testing results. Actran is the only simulation tool able to accurately model the main physical phenomena for engine nacelle radiation."

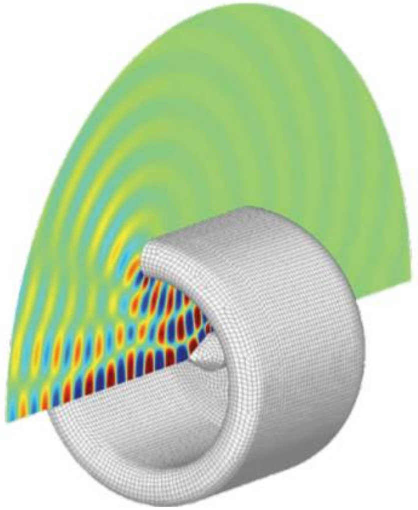
Jean-Yves Suratteau
Head of Numerical Methods, Acoustics & Environment Dept., Airbus

Challenge

- To reduce jet noise-to-noise radiation by an aircraft at takeoff
- Acoustic liners presented a major design challenge because they addressed many conflicting design requirements

Solution

- Airbus had been working to achieve more accurately simulate noise radiation from the nacelle. Actran was the only simulation tool able to accurately model the main physical phenomena for engine nacelle radiation
- Simulation with Actran and other numerical tools helped reveal the substantial impact of splices on forward fan noise and those simulations were confirmed with physical testing



Benefits

- Actran helped Airbus design and deliver best-in-class acoustic solutions that saved aircraft weight with a huge financial impact for airlines operating Airbus aircraft
- Actran also helped Airbus reduce product development costs by avoiding expensive post-design changes
- Airbus is now in the process of expanding its use by developing new applications such as auxiliary power units for ramp noise and counter-rotating open rotor engines for community, cockpit and cabin interior noise

Honeywell

Turbofan engine noise

AIAA 2010-3824, Optimization of a Seamless Inlet Liner Using an Empirically Validated Prediction Method, B.Shuster, L. Lieber - Honeywell Aerospace. A.Vavalle - GKN Aerospace.

Challenge

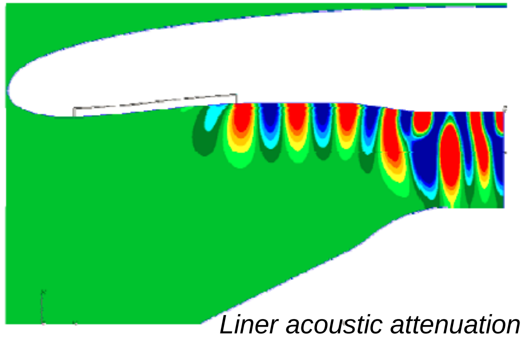
- Aircraft nacelles aim at reducing the noise generated by turbofan engines.
- To this aim reliable acoustic simulation is essential for optimizing the acoustic liner in different flow conditions

Solution

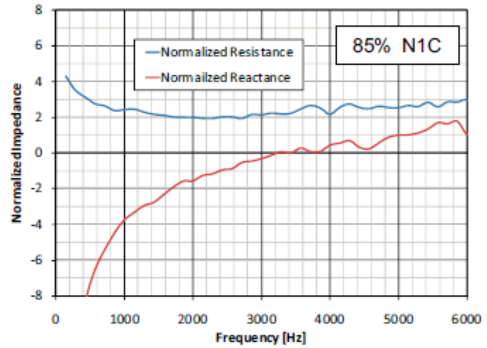
- Actran TM enables the modeling of turbofan engines by taking into account the mean flow both in the near field and far field, the liner effects and complex noise sources at the fan.

Benefits

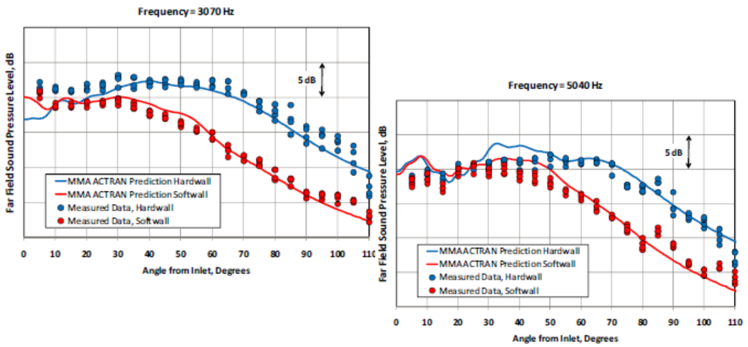
- Actran TM is experimentally validated against ground test results with an excellent correlation.
- The simulation takes directly in input the measured noise source and the liner admittance in presence of grazing flow



Liner acoustic attenuation



Measured Liner impedance



Far-field experimental correlation

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EMBRAER

Repair effect on aircraft nacelle

ICSV 2015, Numerical Investigations Of Noise Radiation From A Turbo-fan Engine In The Presence Of Hard Patches With Different Geometries, O. Acosta, A. da Silva, J. A. Cordioli, D. Reis

Challenge

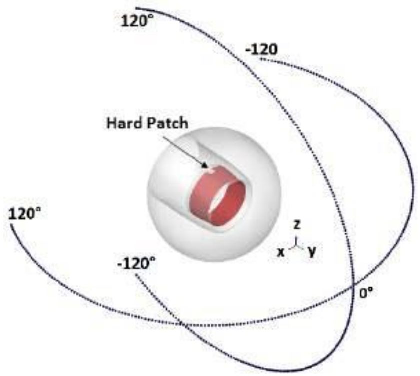
- During engine life-cycle, repairs made in the nacelle liner can result in hard patches acting as discontinuities.
- These discontinuities reduce the acoustic efficiency.

Solution

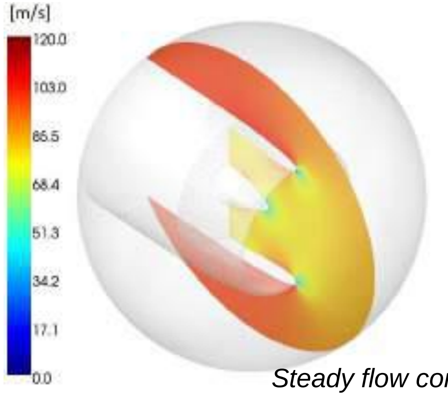
- The compressible flow solver in Actran is used to generate a mean flow to be used in the acoustic model
- Actran TM is used to propagate the noise of the fan in the presence of the flow, taking into account the liner and the hard patches

Benefits

- Actran helped evaluating the acoustic efficiency of liners with different hard patches at different flight conditions.
- At the approach condition the patches have a limited effect, on the contrary at the take-off condition, the hard patches significantly decrease the acoustic efficiency.



Virtual microphones in Actran



Steady flow computed by Actran

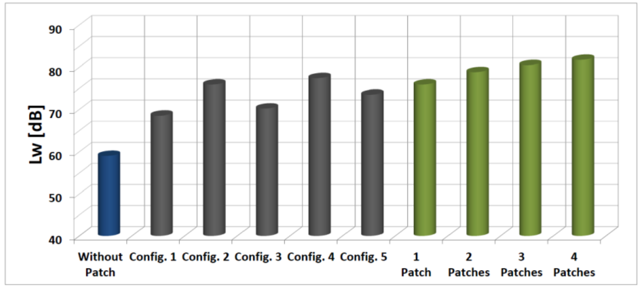


Figure 7: Sound power level. Liner discontinuities. Mode 14.1.

Acoustic power level, for mode (14,1) for several configurations

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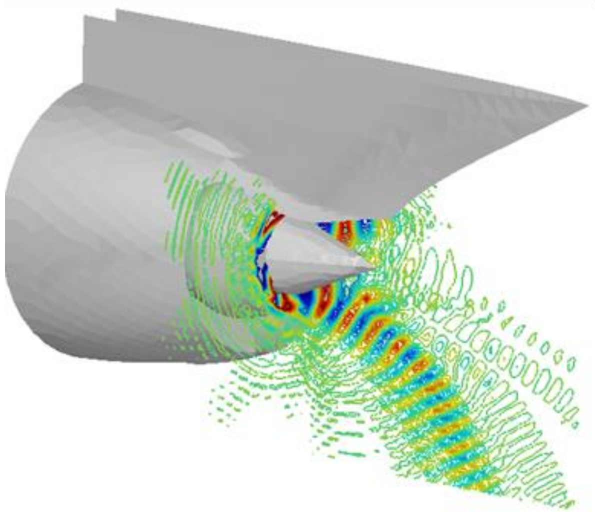
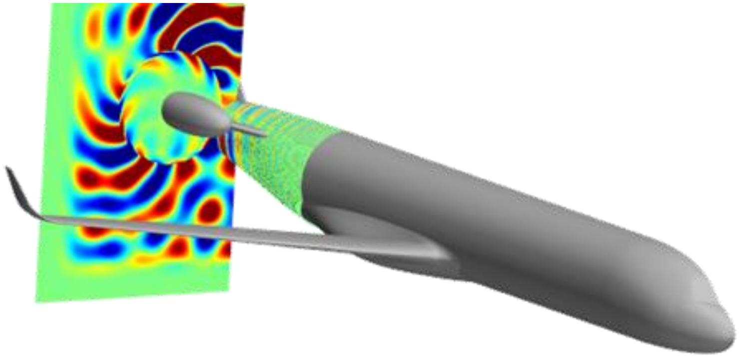
Acoustic propagation for Exhaust and CROR Engines

Acoustic propagation for Exhaust and CROR Engines

- Noise due to:
  - Fan: at idle power / Turbulent jet : at high power
  - Turbine and compressor are minor sources
- How to reduce noise:
  - Use acoustic treatments (liners)
- With Actran you can:
  - Model fan noise source thanks to duct modes
  - Take into account the noise due to open-rotor
  - Import a steady RANS flow from many formats
  - Model acoustic liners with admittance condition or with cavities and perforated plates
  - Propagate the noise in far field thanks to buffer zones (NRBC*) + FWH surface (propagation)
  - Model jet noise and combustion noise



Turbofan exhaust noise with the pylon and bifurcations

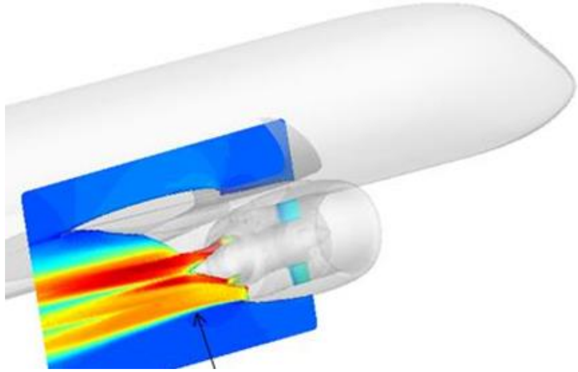


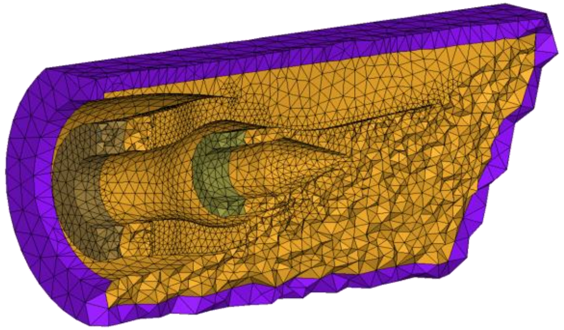
CROR engines noise propagated till fuselage

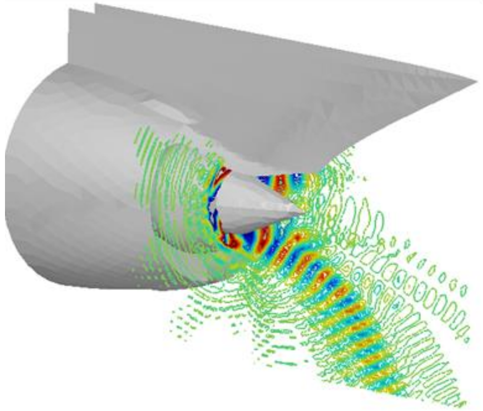
Process for Exhaust Acoustic Radiation

Acoustic model

- Acoustic Liners
- Modal basis
- Buffer zones (NRBC)
- Interpolation of the mean flow







Mean Flow

- RANS Flow imported from an external code

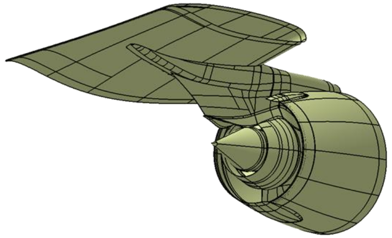
Acoustic Results

- SPL in Near and far field in presence of mean flow
- Radiated and absorbed power

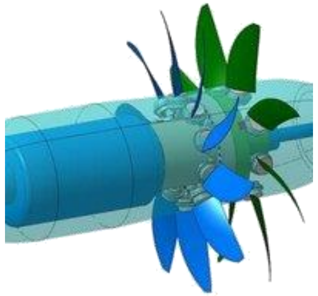
Ingredients for Exhaust / CROR Noise modeling

- Physical Challenges:
  - High flow gradients to take into account in acoustic propagation
  - Supersonic flow
  - Acoustic liners to model and optimize
  - Non-axisymmetric geometry (pylons, bifurcations)
  - Impact of the surrounding environment: wings, fuselage

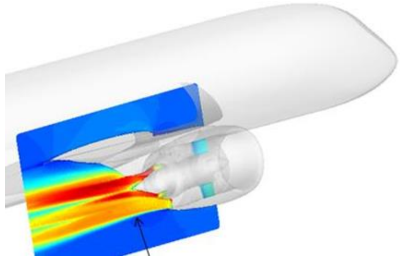
- Solution offered by FFT: Actran DGM
  - Solves the Linearized Euler Equation in time domain with the Discontinuous Galerkin Method (DGM)
  - Allows the propagation in a supersonic flow
  - Aeroacoustic sources around CROR engines / turbofan jets
    ➔calculated on surfaces with Thompson boundary condition



Exhaust of Turbofan



CROR engine



Flow with strong shear layers

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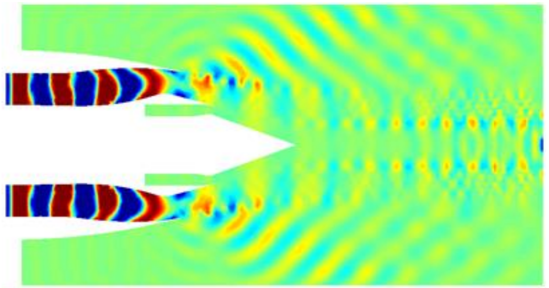
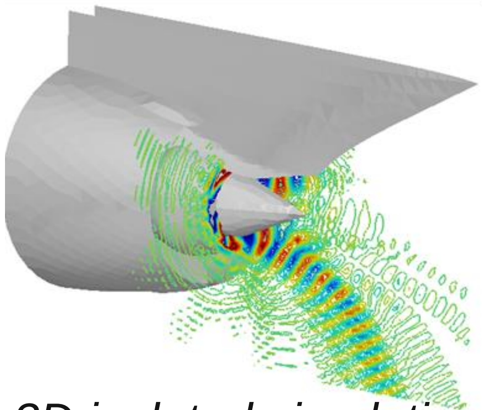
History of the Exhaust Noise at Airbus

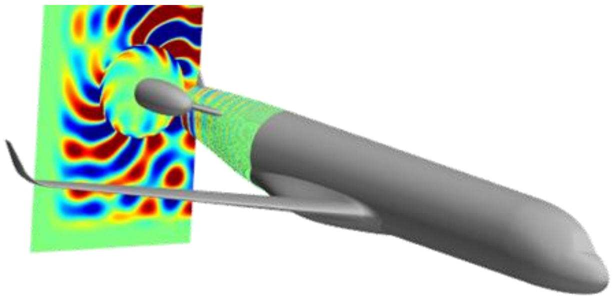
- Actran DGM is used at Airbus since 6 years in R&D context
- Actran DGM was developed and validated in the MESSIAEN & TURNEX EU founded projects
- Actran DGM is well suited for simulating the acoustic propagation in complex sheared flows



2D axisymmetric simulation



3D isolated simulation



CROR Near-field Noise

Time

FFT acoustic conference 2014, Simulation of installation effects of aircraft engine rearward fan noise with Actran/DGM, J-Y. Suratteau

AIRBUS

CROR engine noise

FFT acoustic conference 2014, Simulation of installation effects of aircraft engine rearward fan noise with Actran DGM, J-Y. Suratteau

Challenge

- Evaluate the noise radiated by a new engine generation, the Counter Rotating Open Rotors (CROR).

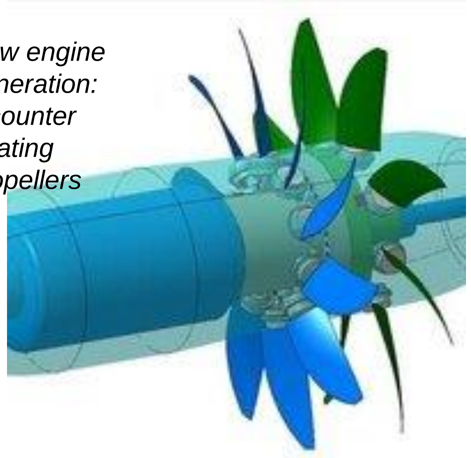
Solution

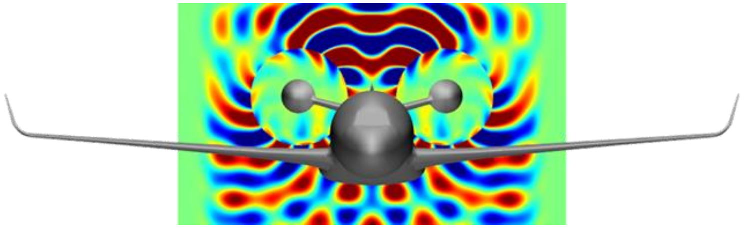
- Actran is used to calculate the aeroacoustic behavior with sources from unsteady CFD results
- Modeling of the refraction of acoustic waves due to the boundary layer on the fuselage

Benefits

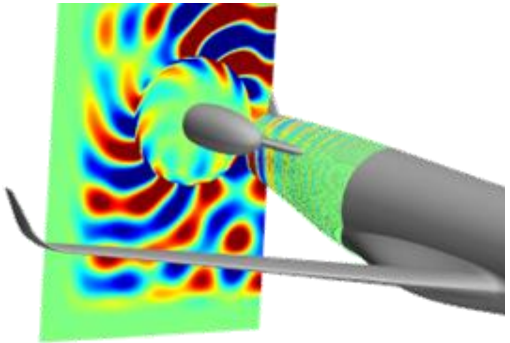
- Addressing industrial aeroacoustic problems for large domains and complex flow

New engine generation: 2 counter rotating propellers





Near field noise due to a CROR engine



CROR engine propagation over the fuselage taking into account the boundary layer

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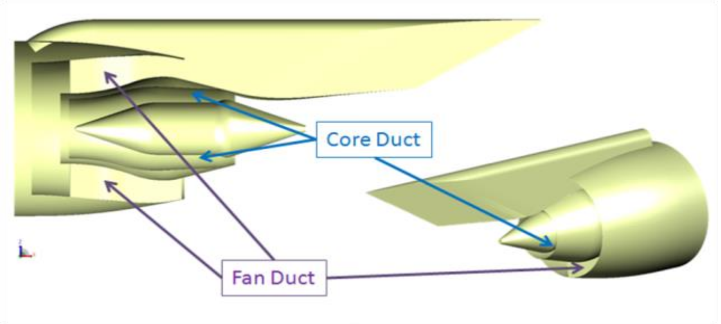
Exhaust Noise of Turbofan Engine with 3D Effects

AIAA Conference 2009, Exhaust noise prediction of realistic 3D lined turbofans submitted to strong shear layers, FFT: R. Leneveu, B. Schiltz, Airbus France: Y. Druon, A. Mosson,

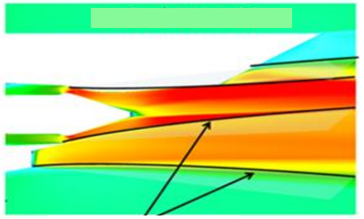
Challenge

- Take into account 3D effect of the turbofan geometry for exhaust noise prediction in take-off condition. Indeed, the geometry of the fan duct is not symmetric due to bifurcations and the pylon.

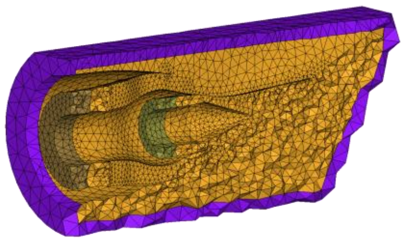


Solution

- Actran DGM is used to perform calculation of a full 3D model.
- Complex flows computed with steady RANS are taken into account including the propagation through shear layers.
- Acoustic treatments are modeled with a surface admittance.

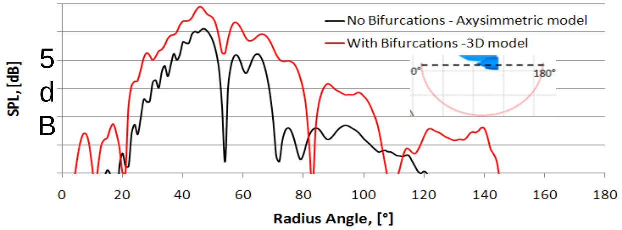


Shear layers



Benefits

- Axisymmetric models are not accurate enough. Bifurcations and pylon must be taken into account for correct prediction.



AIRBUS

Turbofan engine noise: installation effects

OPENAIR European project

FFT acoustic conference 2014, Simulation of installation effects of aircraft engine rearward fan noise with Actran/DGM, J-Y. Suratteau

Challenge

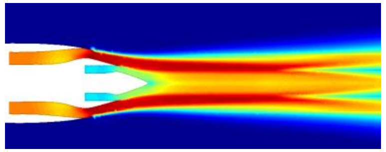
- This project aimed at characterizing aircraft exhaust noise of a 1/10 mock-up, in static condition.
- To this aim, both specific measurement techniques and validated acoustic simulations are used.

Solution

- After a validation on an isolated case, Actran calculated:
  - wing shielding effect on top
  - non-symmetric installation effects on the ground
  - jet blockage effect on the flyover arc
  - interference pattern (reflection and diffraction)

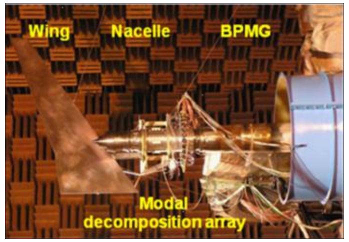
Benefits

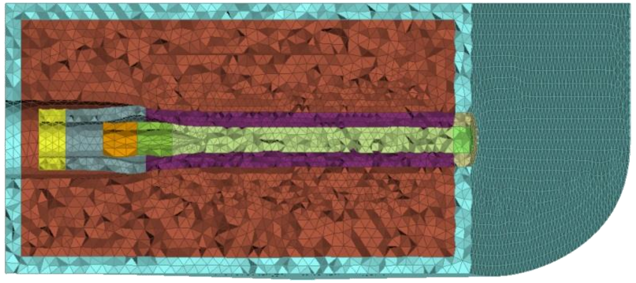
- Actran DGM is able to predict accurately installation effects for aircraft engine exhaust noise
- Good correlation between measurements and simulation is achieved when evaluating very complex acoustic phenomena



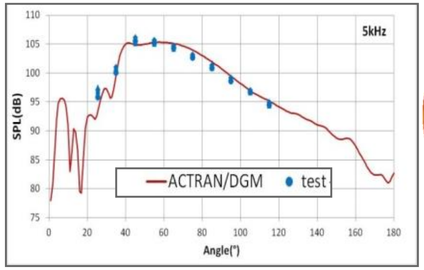
Mean flow imported in Actran: RANS Flow

Experimental set-up

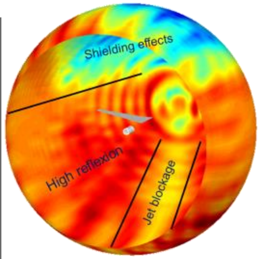




Acoustic mesh use for the Actran model



Actran Vs tests on isolated case (without wing)



Pressure map calculated by Actran around the engine and the wing

HEXAGON

Installation effects of aircraft rear fan noise

AIAA Conference 2014, Simulation of Installation Effects of Aircraft Engine Rear Fan Noise with ACTRAN/DGM, A. Mosson, D. Binet, J. Caprile

Challenge

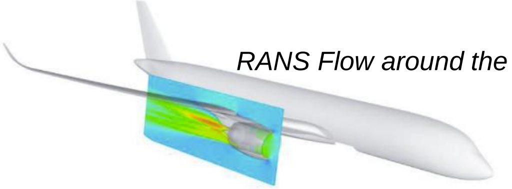
- Simulations of large-scale domains including flow effects such as the acoustic radiation from turbofan engine installed on the aircraft

Solution

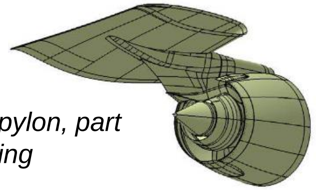
- Actran DGM is used for computing the far field noise taking account of shapes of engine, wing and pylon.
- The acoustic propagation accounts for the mean flow computed by steady RANS simulations.

Benefits

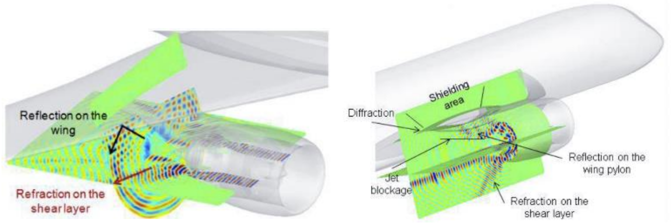
- Good correlation with measurements on a canonical test-case (gaps of 1dB).
- Actran DGM shows a very good computational efficiency and can be used in an industrial context.



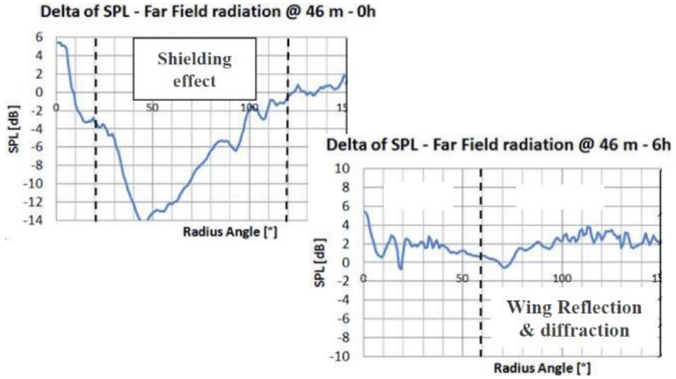
RANS Flow around the engine



Model: engine, pylon, part of the wing

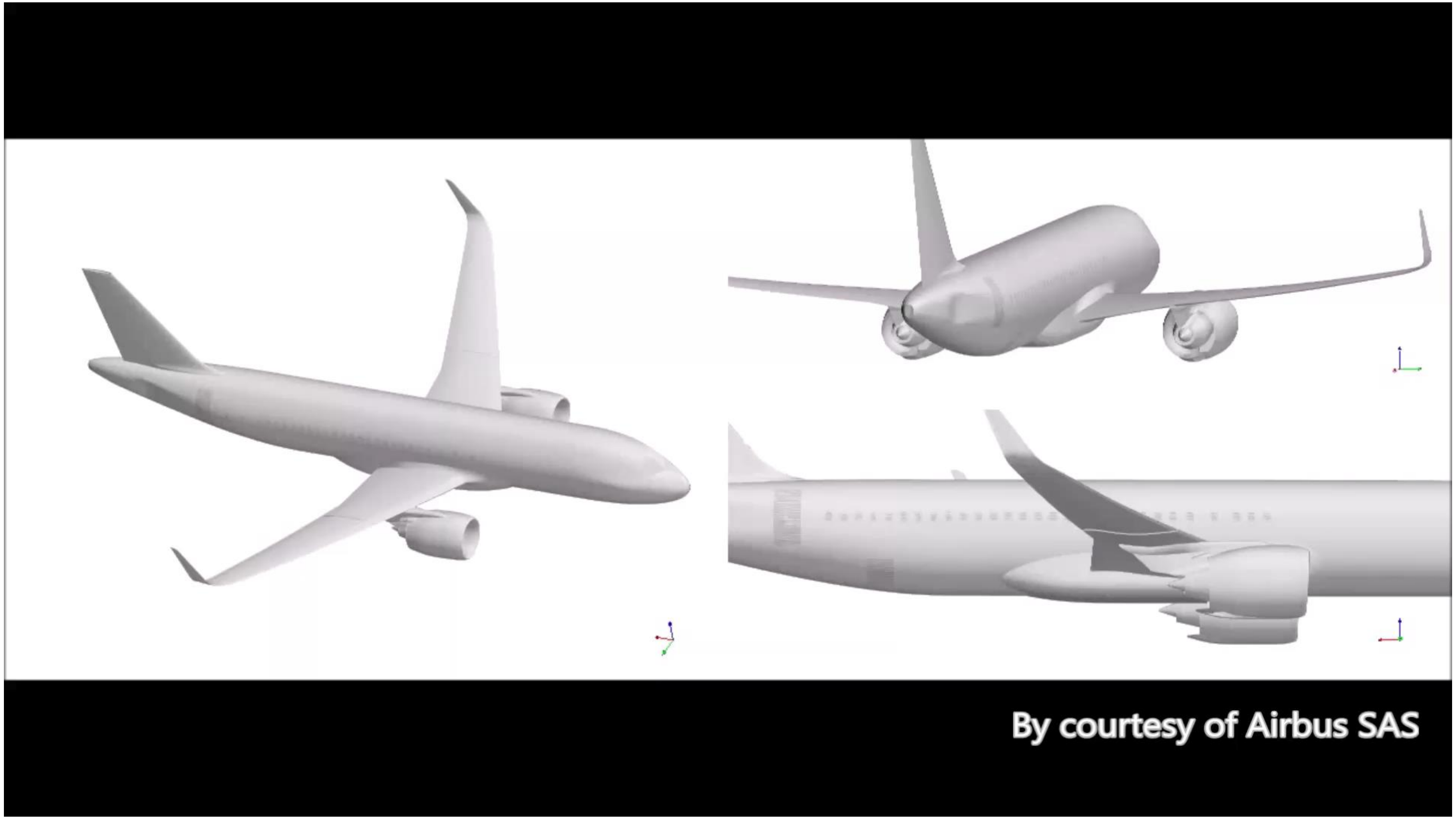


Main acoustic phenomena to simulate



Installation effect (wing effect) above the wing (0H) and below the wing (6H)

Airbus - Installation Effects on an industrial problem



Ramp Noise

Ramp Noise and Regulations

- Ramp Noise: Noise induced by the aircraft on the ground

- Sources of ramp noise
  - APU (Auxiliary Power Unit): the main source especially for exterior noise
  - ECS (Environmental Control System)
- This level of noise is regulated: by each airport following the ICAO (International Civil Aviation Organization) recommendations
- If this level is too high:
  - the APU and the ECS must be switched off
  - the airline must hook to (and pay for) ground electric and conditioned air supply

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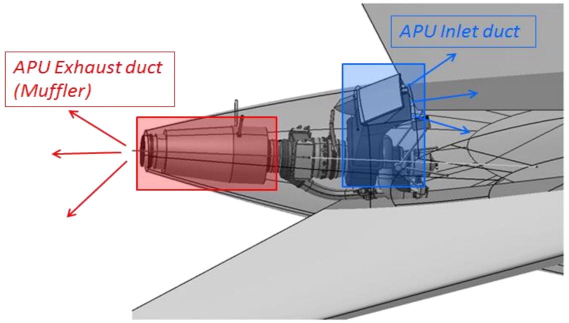
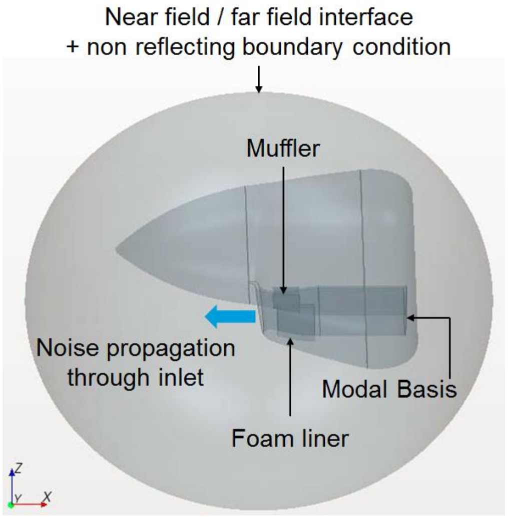
Ramp Noise

- Noise due to:
  - APU (Auxiliary Power Unit)
  - Turbopropeller air intake
- How to reduce noise:
  - Using acoustic treatments
- With Actran you can: (like for air intake)
  - Model easily the turbomachinery noise source thanks to duct modes (analytical or numerical)
  - Perform convected acoustic propagation
  - Use the Actran potential flow solver or import mean flows from many CFD formats
  - Model acoustic liners with admittance condition, cavities and perforated plates
  - Evaluate the TL, the reflected power, the dissipated power and the radiated power



APU: air intake and exhaust



Turbopropeller air intake noise modeling

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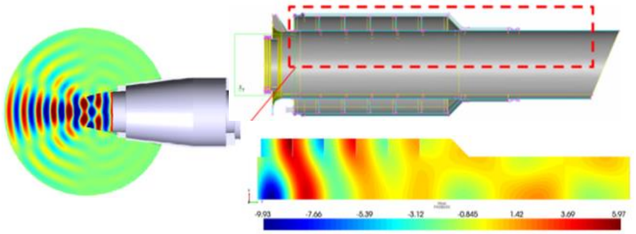
Aircraft composite tail cone

ADVITAC European project

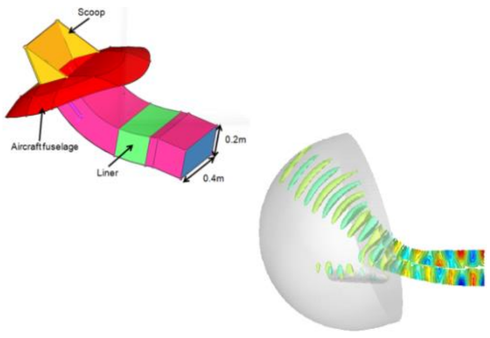
Challenge

- Reducing aircraft ramp noise coming from tail cone, the rear part of aircraft containing the Auxiliary Power Unit (APU)



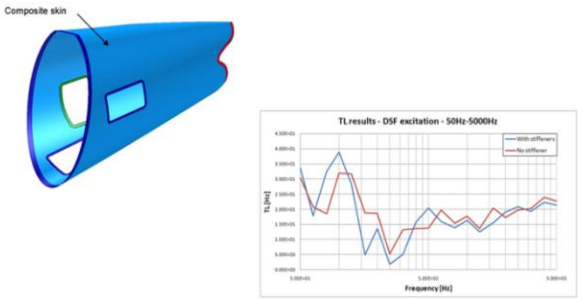
Solution

- Actran TM: APU noise radiation in presence of complex flow, splitters, scoop, porous material, perforated sheet and cavities.
- Actran Vibroacoustics: composite structure modeling, with shell, omega-shaped stiffeners and diffuse sound field excitation on applied on the fuselage.



Benefits

- Reduce APU inlet and exhaust radiation
- Improve tail cone muffler insulation in order to meet the ICAO certification standards



APU composite tail cone – Acoustic transmission losses

Honeywell

Turbo propeller inlet

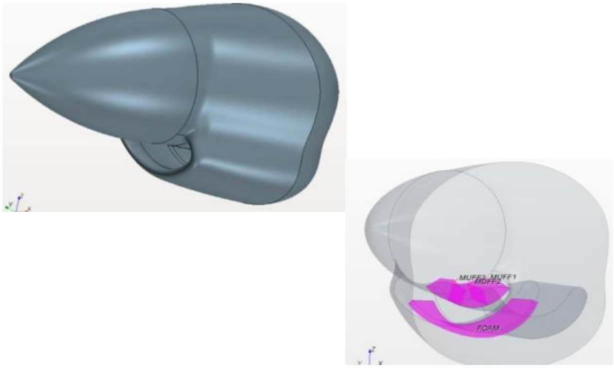
FFT Acoustic Conference 2014, ACTRAN/TM Analyses of Multiple Acoustic Treatment Concepts in the Inlet Duct of a Turboprop Engine, Lysbeth S. Lieber

Challenge

- Turboprop engines suffer from ramp noise at the ground idle operating condition, which affected by high broadband noise levels

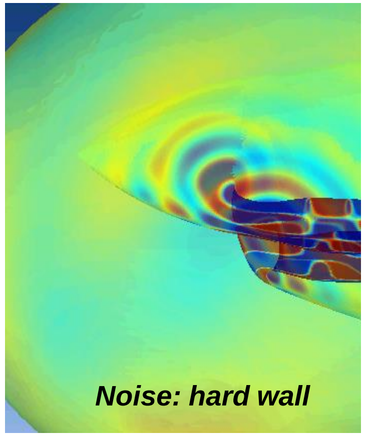
Solution

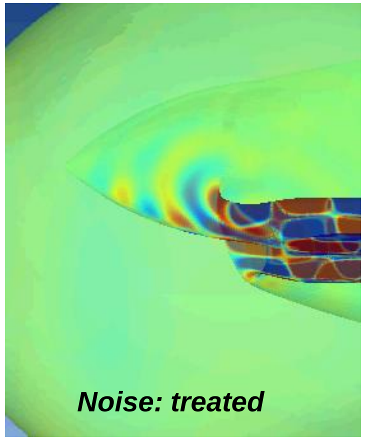
- Actran TM for modeling turbomachinery noise sources of an arbitrary shaped duct
- A muffler is modeled with 3 cavities on the top duct wall covered by a perforated plate. A foam is positioned on the bottom wall covered by a perforated plate.



Benefits

- Prediction of the transmission loss index for many configurations to determine the optimal treatment





Aircraft APU noise evaluation

Challenge

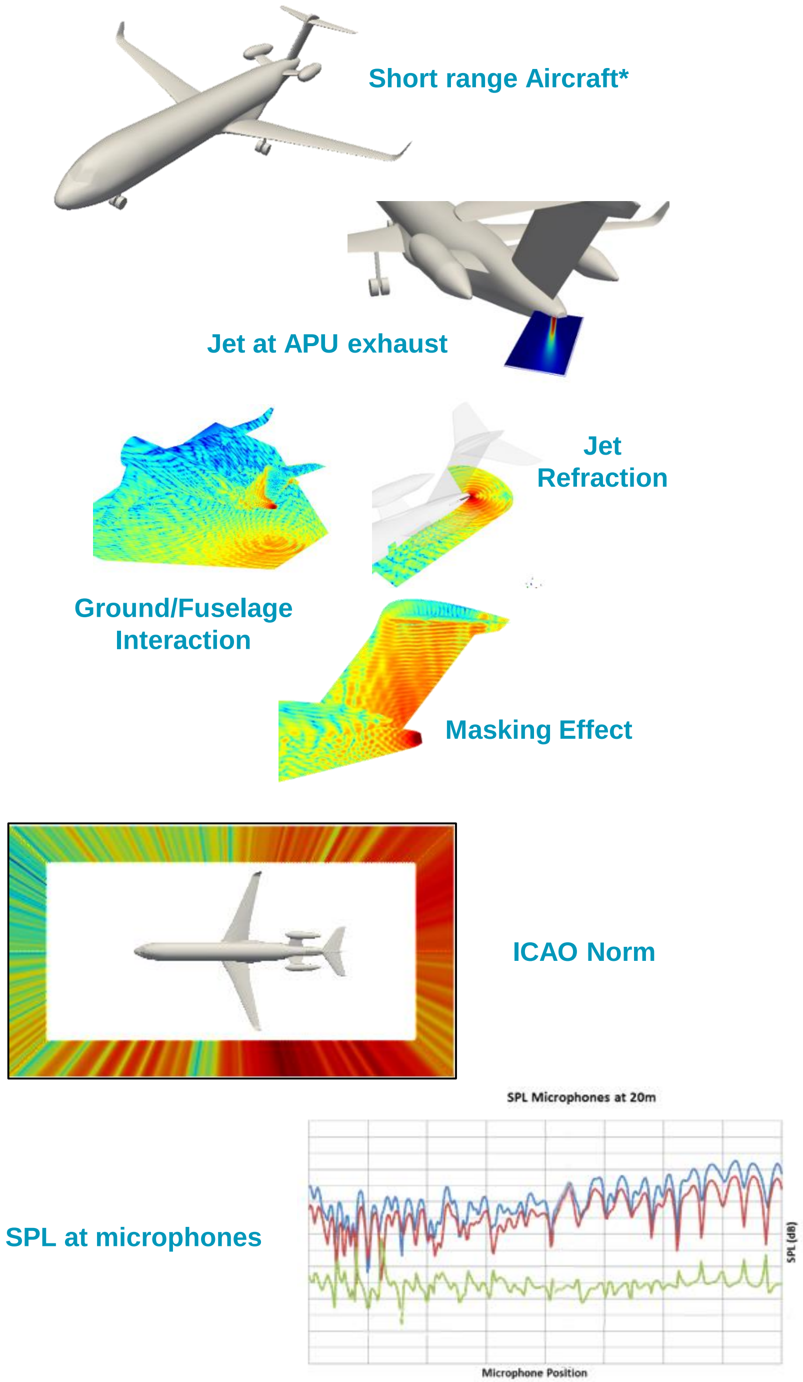
- Prediction of the noise due to Aircraft Auxiliary Power Unit (APU) at ICAO certification points taking into account complex flow due to the APU exhaust and full 3D geometrical effects (e.g. fuselage/ground)

Solution

- Actran DGM solves the acoustic propagation in complex flows (high temperature and velocity gradients) and large acoustic domains.
- This makes it suitable for solving problems like ground-fuselage interaction, masking effects and jet refraction

Benefits

- Actran DGM is a unique tool able to model large acoustic domains (8500 m3).
- The wings, rear fuselage, wheels, ground and VTP/HTP of a single aisle aircraft are modeled.
- Results highlight the strong effect of the flow on the noise prediction at ground.



Aeroacoustic Simulations

Distortion noise, CROR engine, jet noise, combustion noise

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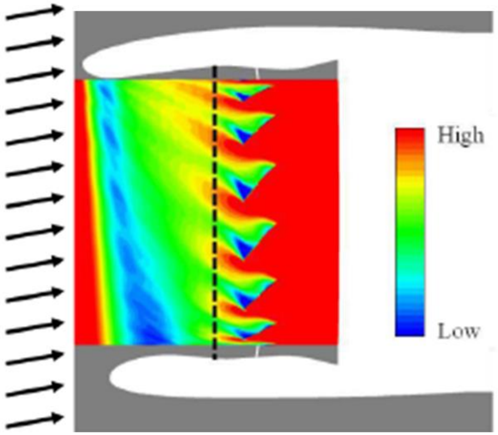
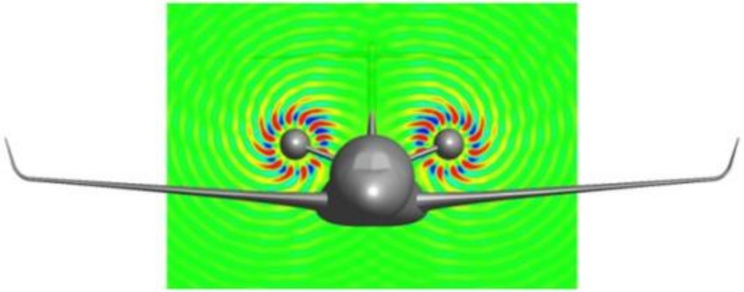
Aeroacoustic simulations

- Noise due to:
  - Distortion noise: caused by short nacelles + flow incidence
  - CROR engines: no nacelle
  - Jet noise: important at high power
  - Combustion noise
- With Actran you can:
  - Read unsteady CFD and compute the corresponding aeroacoustic sources
  - Use one of the three methods available in Actran: Lightill analogy, Möhring analogy and Thompson boundary condition
  - Use acoustic non reflecting boundary conditions
  - Take into account supersonic flows

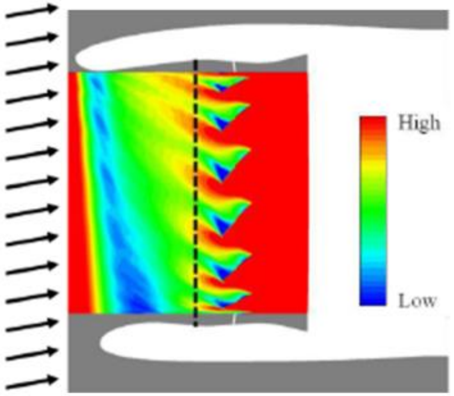


Non uniform load on a short nacelle: →distortion noise



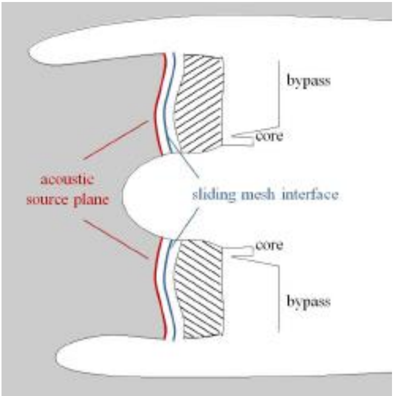
Acoustic radiation of a CROR engine

Typical Process for Aeroacoustic Simulation



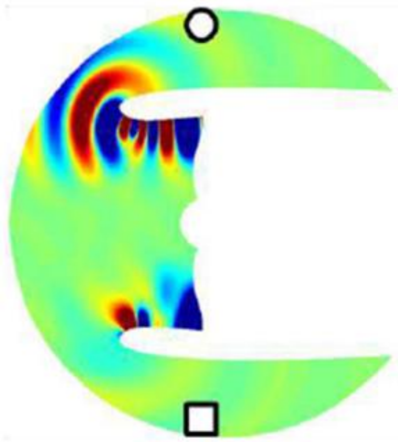
Unsteady CFD

- Extract on a surface the velocity, the density, the pressure (only for Thompson in Actran DGM)



Acoustic model

- Acoustic Liners
- Aeroacoustic sources on a surface (calculated with the iCFD utility)
- Interpolation of the mean flow



Acoustic Results

- SPL in Near and far field

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Calculation of noise sources in Actran

→Lightill and Möhring Analogy

- In Actran:
  - The Lighthill and Möhring analogies are available
  - The Möhring analogy is better suited for Mach numbers > 0.2
    - Viscous stress and entropic variations are neglected
    - Möhring time equation:

$$\frac{\partial}{\partial t}\left[\frac{\rho}{\rho_T^2 c^2}\frac{\partial b}{\partial t}+\frac{\rho v_i}{\rho_T^2 c^2}\frac{\partial b}{\partial x_i}\right]+\frac{\partial}{\partial x_i}\left[\frac{\rho v_i}{\rho_T^2 c^2}\left(\frac{\partial b}{\partial t}+v_i\frac{\partial b}{\partial x_i}\right)-\frac{\rho}{\rho_T^2}\frac{\partial b}{\partial x_i}\right]=R$$

Acoustic Wave Operator Left-Hand Side (LHS)

Source Terms Right-Hand Side (RHS)

$$R=-\frac{\partial}{\partial x_i}\left[\frac{\rho}{\rho_T}(\vec{v}\times\vec{\omega})_i-\cancel{\frac{\partial \tau_{ij}}{\partial x_j}}\right]+\cancel{\frac{\partial}{\partial x_i}\left[\frac{v_i}{\rho_T}\left(\frac{\partial \rho}{\partial s}\right)_p\frac{\partial s}{\partial t}-\frac{\rho T}{\rho_T}\frac{\partial s}{\partial x_i}\right]+\frac{\partial}{\partial t}\left[\frac{1}{\rho_T}\left(\frac{\partial \rho}{\partial s}\right)_p\frac{\partial s}{\partial t}-\frac{\rho v_i}{\rho_T^2}\frac{\partial \rho_T}{\partial x_i}\right]}$$

<u>Sources due to:</u>
vorticity
viscous stress (neglected)
→Turbulent noise

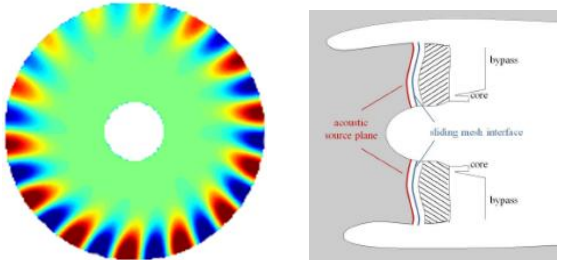
<u>Sources due to:</u>
entropic variations
(neglected but could be implemented)
→ Direct Combustion noise

Turbofan inlet distortion noise

AIAA Conference 2014,Turbofan Inlet Distortion Noise Prediction with a Hybrid CFD-CAA Approach , J. Winkler, C. A. Reimann, R. Reba, J. Gilson

Challenge

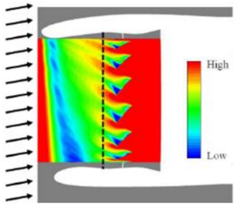
- In short nacelle designs, the noise is generated by the interaction between fan blades and the mean flow (distortion noise).



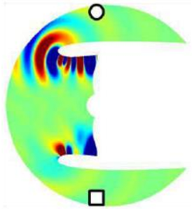
Source plane: acoustic pressure

Solution

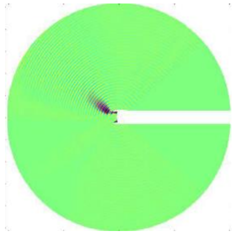
- Actran's Möhring analogy is coupled with a URANS CFD solver for computing and the aeroacoustic sources at the fan face where the surface is non-planar and the flow speed is high (above M=0.3).



Unsteady CFD: velocity



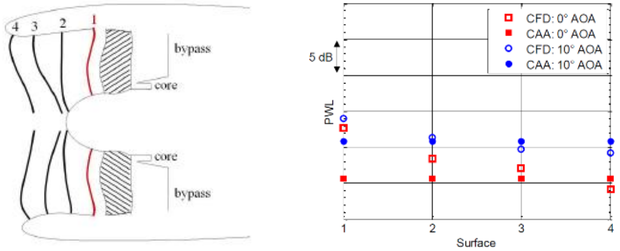
Near field acoustic pressure



Far field acoustic pressure

Benefits

- Actran offers an alternative method to duct modes which are applicable only to planar surfaces with uniform mean flow
- Compared to the full CFD method, Actran offers both better efficiency (hybrid aero-acoustic method) and better accuracy (no acoustic dissipation or reflections that impact the acoustic pressure quantity)



CFD vs Actran: power level at different surfaces

HEXAGON

CROR engine noise

FFT acoustic conference 2014, Simulation of installation effects of aircraft engine rearward fan noise with Actran DGM, J-Y. Suratteau

Challenge

- Evaluate the noise radiated by a new engine generation, the Counter Rotating Open Rotors (CROR).

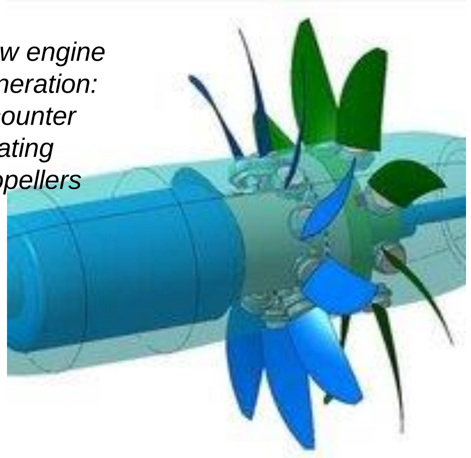
Solution

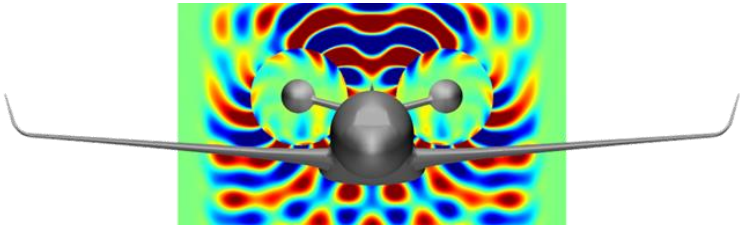
- Actran is used to calculate the aeroacoustic behavior with sources from unsteady CFD results
- Modeling of the refraction of acoustic waves due to the boundary layer on the fuselage

Benefits

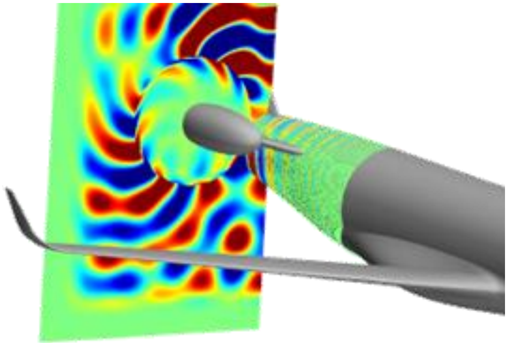
- Addressing industrial aeroacoustic problems for large domains and complex flow

New engine generation: 2 counter rotating propellers

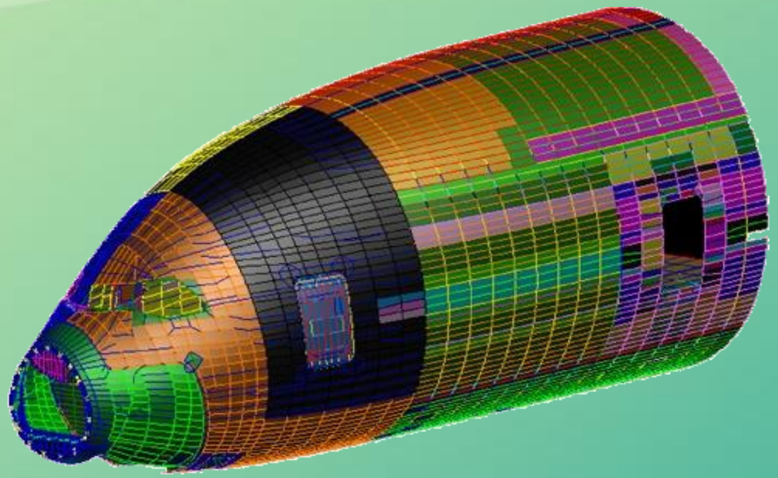




Near field noise due to a CROR engine



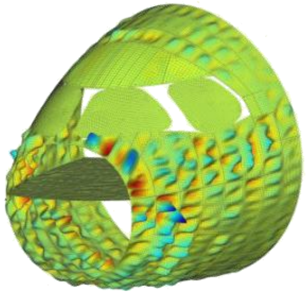
CROR engine propagation over the fuselage taking into account the boundary layer



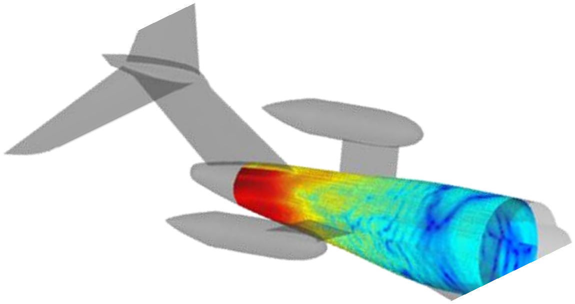
Interior Noise

Interior Noise

- Noise due to:
  - Turbulent boundary layer on the fuselage
  - Engines (intake, exhaust, CROR engines)
  - Air conditioning
- How to reduce noise:
  - Change material properties
  - Use absorbing materials
- With Actran you can:
  - Evaluate the interior noise due to engine noise
  - Calculate noise reduction index of fuselage
  - Predict the TL of air ducts and windshields
  - Apply typical loads: TBL and modal ducts
  - Model structures, air and absorbing materials
  - Import Nastran structural model in Actran

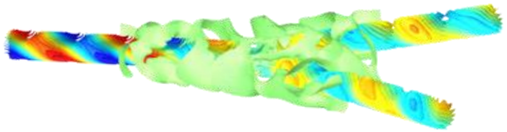


Cockpit vibrations
-
due to the turbulent boundary layer



Load the fuselage with pressure field

(in this case due to CROR engines)



Noise propagation in air conditioning ducts and duct structure radiation

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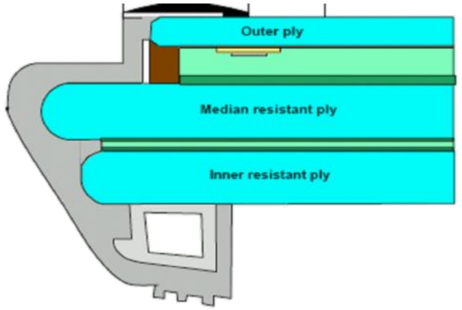
Windshield Transmission Loss

Challenge

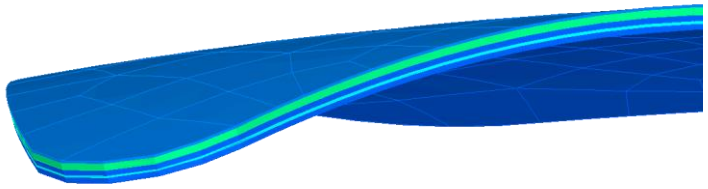
- The windshield vibrates due to the turbulent air flow on its surface. This vibration produce noise that directly radiates on the pilot and co-pilot . The goal is thus to reduce the noise transmission through the windshield.



Windshield model: frame, seals, several layers

Solution

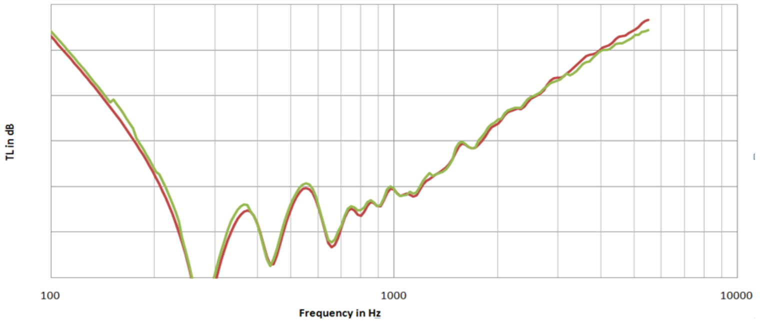
- Actran is used to model the complete windshield with all the plies and the frame. This model is loaded with a turbulent boundary layer, thanks to a random excitation process that does nor required an unsteady flow.



Several windshield layers modeled in Actran

Benefits

- The Transmission Loss is computed up to 8kHz on different windshield configurations to predict the best one.



Windshield Transmission Loss (dB)

HEXAGON

Aircraft cabin noise

Challenge

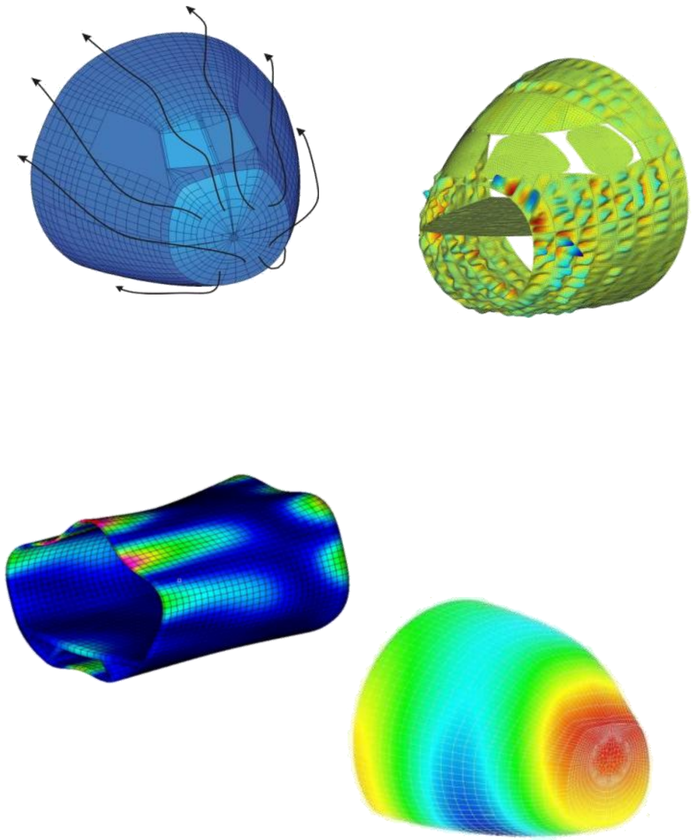
- Aircraft noise is not only important in terms of the exterior but also in terms of the interior of the aircraft.
- It is important to reduce the noise transmission through an aircraft fuselage or cockpit in order to increase comfort for passengers and crew alike

Solution

- Actran’s vibroacoustics module is able to account for the complex structure (composite fuselage and multi-layered windows), the interior cavity, the insulation materials and the pressurization effect
- It provides a holistic view on the transmission of noise to the cabin

Benefits

- Acoustic simulation enables Airbus to understand the effects of various parameters such as the excitation type, the presence of the floor and the variable thickness on the interior noise levels



Aircraft interior noise

CRESCENDO European project

Challenge

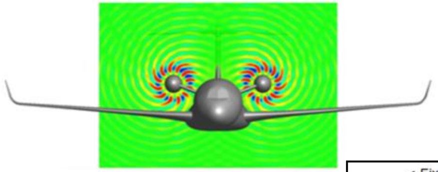
- New engine concepts can provide better noise performance to align with new regulations
- The **Counter-Rotating Open Rotor (CROR)** is a very promising concept where two counter-rotating rotors are used to propel the aircraft
- Considering noise issues at design phase of an aircraft equipped with CROR engines is important for advancing the concept

Solution

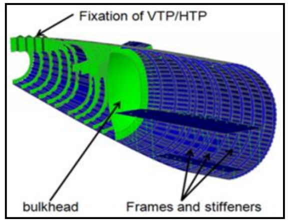
- SimXpert: Nastran structure modeling
- Nas2Act: conversion of Nastran model into an Actran model
- Actran: enrichment of the Actran model with fluid, foam materials and non-uniform pressure excitation on the fuselage

Benefits

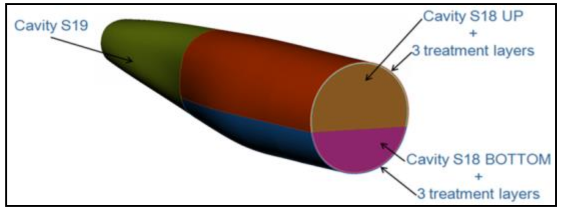
- Compute **sound pressure level in the cabin** due to airborne excitation of **synchronous** and **asynchronous engines**



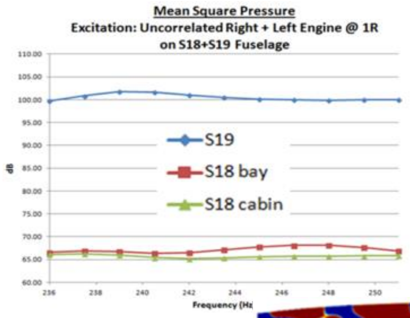
Acoustic Pressure fields

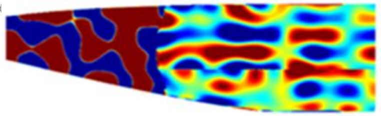


APU composite tail cone – Acoustic transmission losses



APU composite tail cone – Acoustic transmission losses





AIRBUS

Air supply ducts

Noise-Con 2008, Design sensitivity analysis on low-frequency noise radiation of air supply ducts, P. Neple, R. Foret, A. Ramonda

Challenge

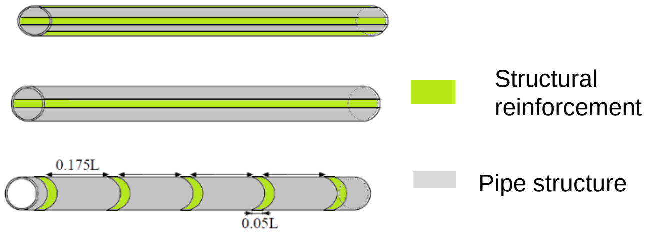
- The air conditioning system of an aircraft is one potential source of interior noise.
- Duct performance is driven by the coupling of the structural modes and the acoustic plane wave: it is needed to study the vibroacoustic propagation through the HVAC duct.

Solution

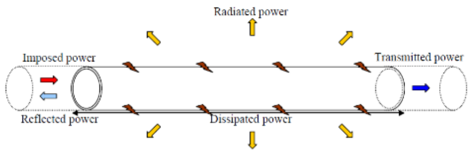
- Actran simulates the noise propagation within the pipe, taking account of the pipe structure.
- Actran is used to compute:
  - the transmitted power through the air outlet
  - the radiated power by pipe structure

Benefits

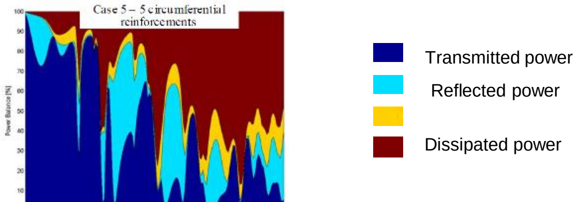
- Coupled vibro-acoustic behavior of the HVAC duct
- Ranking of the different configurations
- The analysis allowed to identify improvements of acoustic transmission loss for the pipes



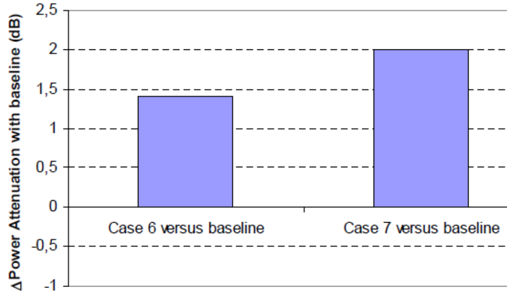
Some studied pipes configurations



Power balance with Actran



Power balance with Actran



Improvement of acoustic transmission losses between inlet and outlet

HEXAGON

LIEBHERR

SAFRAN

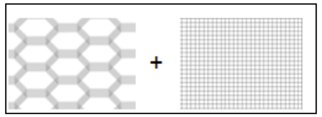
Aircraft ventilation system

SOFTAIR French project

Challenge

- Define, propose and validate an optimal acoustic treatment for an air conditioning plenum.
- The validation shall be performed between measurements and calculations in flow and no flow condition on a test bench.

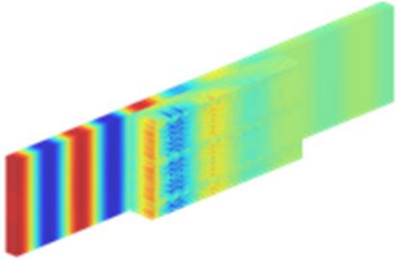
Acoustic treatment modeled in Actran



NIDA and Wiremesh definition in Actran

Solution

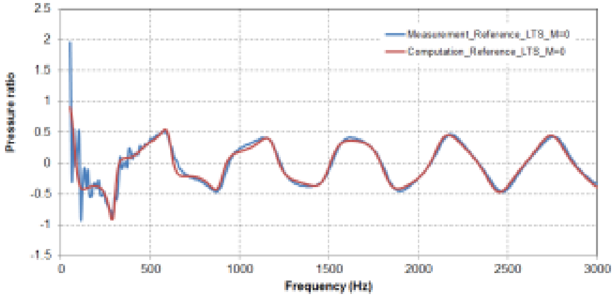
- Actran and Patran have been linked together for defining DOE on several acoustic treatments.



Acoustic pressure calculated in a duct with treatment

Benefits

- Very good correlation with measurements has been achieved.
- Large range of configurations have been simulated to help finding the optimal one.



Pressure ratio (real part) upstream / downstream the treatment:
Measurements Vs Actran

HEXAGON